

Trigonometric functions on the unit circle



1

a $\frac{1}{2}$

b $\frac{\sqrt{3}}{2}$

c $\frac{\sqrt{2}}{2}$

d $\frac{\sqrt{3}}{2}$

e 0

f 1

g 1

h $\frac{1}{\sqrt{3}}$

2

a $-\frac{1}{2}$

b $-\frac{1}{2}$

c -1

d $-\frac{1}{2}$

e $\frac{\sqrt{3}}{2}$

f -1

g $-\frac{1}{\sqrt{3}}$

h $-\frac{1}{2}$

3

a $\frac{1}{y}$

b $\frac{1}{x}$

c $\frac{x}{y}$

4

a $\frac{2}{\sqrt{3}}$

b 1

c $-\frac{2}{\sqrt{3}}$

d 2

e $-\frac{\sqrt{2}}{-\sqrt{2}}$

f $-\sqrt{2}$

g $-\sqrt{2}$

h 1

5

a $\frac{2}{\sqrt{3}}$

b $\frac{2}{\sqrt{3}}$

c -2

d $-\frac{2\sqrt{3}}{3}$

e $\sqrt{3}$

f $\sqrt{2}$

g $-\frac{2\sqrt{3}}{3}$

h $\sqrt{3}$

6

a i Quadrants 1 and 2

ii Quadrants 3 and 4

b i Quadrants 1 and 4

ii Quadrants 2 and 3

c i Quadrants 1 and 3

ii Quadrants 2 and 4

7



a

i $\frac{\sqrt{3}}{2}$ ii $\frac{1}{2}$ iii $\sqrt{3}$ iv $\frac{2}{\sqrt{3}}$
 v 2 vi $\frac{1}{\sqrt{3}}$

b

i $\frac{\sqrt{21}}{5}$ ii $-\frac{2}{5}$ iii $-\frac{\sqrt{21}}{2}$ iv $\frac{5}{\sqrt{21}}$
 v $-\frac{5}{2}$ vi $-\frac{2}{\sqrt{21}}$

c

i $\frac{-1}{\sqrt{2}}$ ii $\frac{1}{\sqrt{2}}$ iii -1 iv $-\sqrt{2}$
 v $\sqrt{2}$ vi -1

8

a

i $\frac{1}{\sqrt{2}}$ ii $\frac{1}{\sqrt{2}}$ iii 1

b

i 135° ii 225° iii 315°

c

i $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ ii $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ iii $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

d

i $\sin 45^\circ$ ii $-\cos 45^\circ$ iii $-\tan 45^\circ$ iv $-\sin 45^\circ$

e

i $\frac{1}{\sqrt{2}}$ ii $-\frac{1}{\sqrt{2}}$ iii -1 iv $-\frac{1}{\sqrt{2}}$

9

a No solution provided.

b No solution provided.

c No solution provided.

d No solution provided.